

Qijin Xu

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Summary

Lead AI/ML Engineer delivering measurable impact via production AI systems. Drove 90% support reduction (525% ROI) with RAG for 50k+ users, and 50% labor reduction with LLM agents (LangGraph), drawing on full-stack expertise from data systems (Spark, Kafka), classic ML (321% fraud lift) to modern LLMs.

Technical Skills

Languages: Python, Java, Go, SQL(TSQL, PostgreSQL), JavaScript, C++

AI/ML: PyTorch, LangGraph, Retrieval-Augmented Generation (RAG), agent workflows / agents, Model Context Protocol (MCP), Large Language Models (LLMs), FAISS, Pinecone, OpenAI API, vLLM, JAX

Data/Streaming: Apache Spark, Kafka, Airflow, Data Lineage (OpenLineage), Redis

Cloud/Infra and Systems: Azure, GCP, AWS, Docker, Kubernetes, Terraform, Microservices, gRPC, CI/CD

Experience

CGI Inc.

Toronto, ON

Lead AI/ML Engineer (Tech Lead), AI Center of Excellence

Jan 2025 - Present

- Lead an AI agent for a proprietary coding language, with a LangGraph dual-state dynamic workflow, vLLM-served LLMs with RAG and EBNF structured outputs, PII filtering, documentation, and business-rule extraction; projected 50% labor reduction, maintain legacy scripts, and help future migration readiness.

Data Architect and Engineering Lead, Senior Consultant

Aug 2023 - Dec 2024

- CGI Impact Award 2024: Architected and shipped an LLM-powered enterprise RAG chatbot (LangChain, Azure/AWS/GCP); achieved approx. 90% autonomous resolution for a 50K+ user base and delivered 525% ROI in 12 months. [PRODUCT PAGE](#), [COVERAGE](#)
- Built an OpenLineage-based lineage tool auto-documenting extensible multi-source flows, with LLM-assisted root-cause analysis suggestions, reduced investigation time across 3 incidents per month.
- Automated AxiomSL rule updates with LLM at RBC; 80% manual cut.

Data Engineer, Consultant

Aug 2021 - Jul 2023

- Managed Canada's largest automobile-insurance policy-history database (2TB big on SQL server), holding 99.9% uptime and less than 200 ms median queries.
- Built production ETL/ELT pipelines for 22+ micro data marts (less than 6 tables each; monthly SLAs) powering finance, analytics and delivery department; cut report runtime from 30 min to 5 min.
- Shipped a machine learning (Gradient-Boost + Logistic Regression) loss-ratio model that lifted recovered fraud losses 321% versus a top-5 bank benchmark, batch-scoring in a Python/T-SQL pipeline.

Elder Laboratory, York University

Toronto, ON

Research Assistant, Computer Vision (part-time; concurrent)

May 2020 - Aug 2021

- Developed AttentiveVision, a computer-vision pipeline for feature extraction on hockey broadcast video in Python (PyTorch, TensorFlow, OpenCV), training a CNN-LSTM stack that reached mAP greater than or equal to 70% on event detection across 18 hours of game video.
- Engineered a real-time player tracking system using a Kalman-filter tracker and OpenPose, delivering less than 100 ms latency at 30 FPS, capturing pose and momentum for live speed & heat-map analytics.
- Created an event-detection and annotation (Goal, Timeout, etc.) model for sport game videos, cutting manual labelling effort and raising dataset reliability to Cohen's kappa greater than or equal to 0.92.

Education

York University

Toronto, ON

Bachelor of Science in Computer Science, Specialized Honours

Sep 2017 - Apr 2020

GPA: 3.9/4.0; First in Class; Dean's List (all semesters)

Certifications

Microsoft Certified: Azure Solutions Architect Expert, Google Cloud Professional Cloud Developer